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Total No. of printed pages = 03

Monsoon, 2024

B. Tech 3rd Semester Examination

SOLID MECHANICS

Course Code: BCE23301T

Full Marks – 50

Time – 2 hours

*The figure in the margin indicates full marks for the questions.***Part A***Answer ALL questions*

Multiple Choice Questions (10 x 1 mark = 10 marks)

1. a) Modulus of elasticity is the ratio of:
 - (i) Lateral stress to longitudinal stress
 - (ii) Deformation to original length
 - (iii) Longitudinal stress to longitudinal strain
 - (iv) Stress to original length
- b) Hook's law holds good up to:
 - (i) Elastic limit
 - (ii) Yield point
 - (iii) Plastic limit
 - (iv) Fracture point
- c) The radius of Mohr's circle is equal to _____.
 - (i) σ_x
 - (ii) τ_{max}
 - (iii) σ_y
 - (iv) $(\sigma_x + \sigma_y)/2$
- d) The relationship between Young's modulus (E), Modulus of rigidity (C) and Bulk modulus (K) is given by
 - (i) $E=9CK/(C+3K)$
 - (ii) $E=9CK/(2C+3K)$
 - (iii) $E=9CK/(3C+K)$
 - (iv) $E=9CK/(C-3K)$
- e) A cantilever beam of span 3m carries a point load 100 N at the free end. The maximum B.M in the beam will be _____.
 - (i) 100 N-m
 - (ii) 300 N-m
 - (iii) 150 N-m
 - (iv) 600 N-m
- f) The rate of change of bending moment is equal to _____.
 - (i) Shear force
 - (ii) Slope
 - (iii) Deflection
 - (iv) None of these
- g) In simply supported beams, deflection is zero at _____.
 - (i) Mid span
 - (ii) Supports
 - (iii) Throughout the span
 - (iv) Point of action of load

h) If the boundary condition for a real beam end is slope, $\theta = 0$ and displacement, $y = 0$, then, the boundary condition for the corresponding conjugate beam is:

(i) $\theta = 0$; $y = 0$

(ii) $y = 0$; $V = 0$

(iii) $M = 0$; $V = 0$

(iv) $M = 0$; $\theta = 0$

where, M is the bending moment and V is the shear force of the conjugate beam.

i) Polar moment of inertia of a solid shaft of diameter (d) is

(i) $\frac{\pi d^3}{32}$

(ii) $\frac{\pi d^4}{32}$

(iii) $\frac{\pi d^3}{16}$

(iv) $\frac{\pi d^4}{16}$

j) If the circumferential stress developed in a pipe 1.5 m diameter and 1.5 cm thickness is 60 N/mm^2 , the value of internal fluid pressure is _____ N/mm^2 .

(i) 0.2

(ii) 1.0

(iii) 1.2

(iv) 3.4

Part B

Answer ANY Two questions

(2 x 10 mark = 20 marks) (Max. 200 words each)

2. (a) A steel bar of 16 mm diameter is subjected to a pull of 30 kN. The measured extension on gauge length of 200 mm is 0.13 mm and change in diameter is 0.002 mm. Calculate –
 - i. Young's Modulus
 - ii. Poisson's ratio
 - iii. Bulk modulus.
- (b) Find the moment of inertia of a T- beam having flange dimension as 300 mm x 20 mm and a web dimension of 20 mm x 200 mm.
3. (a) A beam is simply supported and carries a uniformly distributed load of 50 kN/m run over the whole span. The section of the beam is rectangular having depth as 500 mm. If the maximum stress in the material of the beam is 120 N/mm^2 and moment of inertia of the section is $6 \times 10^8 \text{ mm}^4$, find the span of the beam.
- (b) A rectangular beam 100 mm wide and 200 mm deep is simply supported over a span of 6 m. If shear force at a section of the beam is 5000 N, find the shear stress at a distance of 35 mm above the neutral axis.
4. (a) For a simply supported beam with cross section 150 mm x 300 mm and of length 3 m, is carrying a load of 10 kN at its mid-span. Find the deflection and slope at the mid span of the beam using moment area method. Take $E = 2 \times 10^5 \text{ N/mm}^2$.
- (b) A thin cylinder of internal diameter 1.5 m contains a fluid at internal pressure of 2 N/mm^2 . Determine the maximum thickness of the cylinder if –
 - (i) The longitudinal stress is not to exceed 20 N/mm^2 .
 - (ii) The circumferential stress is not to exceed 40 N/mm^2 .

Part C

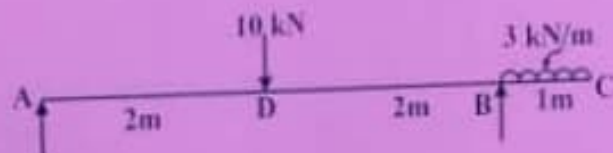
Answer ANY TWO questions

(2 x 10 mark = 20 marks) (Max. 200 words each including subdivisions)

5. (a) A point in an element consists of normal stresses of $\sigma_1 = 200 \text{ N/mm}^2$ tensile and $\sigma_2 = 90 \text{ N/mm}^2$ tensile on two mutually perpendicular planes, together with a shear stress across these planes of 40 N/mm^2 . With analytical formula find the normal stress, shear stress, resultant stress on a plane making an angle 30° with the plane of σ_1 . Find the major and minor principal stress. (5)

(b) For the same element mentioned above, draw the Mohr's circle and determine the normal, tangential and resultant stresses on a plane inclined at 30° to the axis of the minor stress. (5)

6. (a) Draw the shear force and bending moment diagram for the beam shown below and find the point of contraflexure. (6)



b) Find the value of maximum deflection and maximum slope of a cantilever beam of length, L m carrying a load of W kN at a distance of ' a ' m from the fixed end having flexural rigidity, EI . (4)

7. (a) A simply supported beam of length 4 m, is carrying a point load of 6 kN at a distance of 3 m from the left end. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 1 \times 10^8 \text{ mm}^4$ to determine: (5)

(i) slope at left support, (ii) deflection under the load and (iii) maximum deflection.

(b) A hollow circular shaft of diameter ratio $3/8$ (internal to external) is to transmit 375 kW power at 100 rpm. The maximum torque being 20% greater than mean. The shear stress should not exceed 60 N/mm^2 and the twist in the length of 5 m not to exceed 30° . Calculate the external and internal diameters. Assume modulus of rigidity as $0.85 \times 10^5 \text{ N/mm}^2$. (5)